COURSE 2A

C2-03 PROFESSIONAL POWER BI MODELING

BEGINNER-FIRST TEACHING BOOK - RC1

Senior / BI Analyst optional depth path

Grain -> star schema -> relationships -> validation -> measured performance

THIS BOOK IS YOUR TEACHER

Power BI Desktop live-model checks remain learner-PC evidence; this book is self-sufficient for concepts.

This book is your teacher

Use one lesson at a time. New advanced concepts are still taught from first principles. Save a genuine attempt before opening protected review; do not skim ahead just because a topic name looks familiar.

Daily lesson loop

1) Why it matters -> 2) what you learn -> 3) prerequisite -> 4) picture it in your head -> 5) new words -> 6) watch / visual source when useful -> 7) follow with me once -> 8) expected result -> 9) why it worked -> 10) common beginner mistakes -> 11) recovery ladder -> 12) small fresh task -> 13) validate -> 14) teach it back -> 15) protected review + changed retry -> 16) evidence / mastery + exact next route.

Source rule

External videos are companions, not the course authority. If a useful source exists, open the exact link and return to the course task. If an external source would leak a solved capstone or add little value, the lesson explicitly uses an INTERNAL TEACHING ROUTE.

Attempt protection

Do the teacher example once. Then close it and complete the changed task from your own blank workbook/query/report/notebook. Protected review opens only after a saved attempt and is followed by a changed retry.

Mastery standard

You are finished only when you can reproduce the result, validate it, explain why it works, recover from a mistake and save the required evidence.

Contents / Index

Every lesson and actual page number in this RC1 book

Prerequisite and tool map

What is assumed, what is new, and what must be executed in Desktop

ASSUMED

Course 1 basic Power BI navigation/visuals, simple data cleaning and basic relationship awareness. C2-00 grain/metric/validation language. C2-01 Power Query helps but PBM-A8 re-teaches the BI-model architecture from first principles.

TOOLS

Power BI Desktop for hands-on modeling. CSV/XLSX practice package supplied. A spreadsheet app may be used for model-design/QA workbook evidence. No Fabric capacity, Power BI Service subscription or paid cloud tenant is required for C2-03 core mastery.

NEW DEPTH

Semantic contract, fact/dimension roles, conformed star schemas, filter propagation, M:M diagnosis/bridges, role-playing dimensions, governed date table, BI Power Query architecture, model QA and performance fundamentals.

OPTIONAL PATH

C2-03 is optional Senior/BI depth. C2B also teaches modeling at the depth needed for the direct DA-to-DE bridge. C2A must not become a hidden prerequisite for C2B or Course 3.

Practice, review and Gate route

Design -> build -> test filters -> reconcile -> explain -> changed retry

Contract -> Build -> Keys -> Relationships -> Filter test -> Reconcile -> Review -> Retry

LESSON RULE

Read the exact book pages, then build only the matching practice task. Save a genuine model/design attempt, relationship screenshots or notes, and independent controls before protected review.

MINI-LAB

After PBM-A1-A10, build and defend an integrated parcel-delivery semantic model. The case contains mixed grains, a duplicate dimension key, unknown key, date roles and higher-grain targets.

GATES

Gate A is fresh Retail Sales & Returns. Gate B Clinic Encounters & Billing and Gate C Logistics Shipments & Service Targets remain protected full reassessments. Critical model defects block pass.

PBM-A1 - Semantic-model thinking

1. Why it matters

Power BI lets you connect tables quickly, but a model can look tidy while answering the wrong business question. Modeling starts before you draw relationship lines: decide what process, row grain and questions the model must support.

2. What you are learning

Write a semantic-model contract: business process, intended facts, dimensions, fact grain, required questions, measure candidates, security/refresh boundaries and validation controls.

3. Prerequisite / what you already know

Course 1 Power BI basics and C2-00 grain/metric-contract thinking. No advanced DAX is required.

4. Picture it in your head

Mental model

Imagine this as a visible path: Business process -> Row grain -> Questions -> Tables -> Relationships -> Controls. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Semantic model: The organized layer of tables, relationships, metadata and calculations that gives business meaning to source data. Fact grain: What one row in a fact table represents. Model contract: A short specification of what the model must represent, support and validate.

6. Watch / visual source

Model contract + sketch + validation plan + 60-90 second explain-back. VIDEO SUPPORT: 00:43-18:07
https://www.youtube.com/watch?v=4ePNrdxWtY0
https://learn.microsoft.com/en-us/power-bi/guidance/star-schema
After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Business process: online sales and returns. Decision: compare revenue, margin and return rate by product, customer, region and date.
2. Declare FactSales grain = one row per order line. FactReturns grain = one row per return event; do not pretend both facts have the same grain.
3. List dimensions needed for filtering/grouping: Product, Customer, Date and possibly Region if it is governed independently.
4. State the first validation controls: FactSales row count, distinct order-line key, gross/net sales totals, unmatched ProductID/CustomerID and return-to-sale reconciliation.
5. Only after this contract should you decide which relationships belong in Power BI.

8. What you should see / be able to say

Expected result

A one-page model contract where a reviewer can predict the intended star, fact grains and controls before seeing the PBIX.

9. Why it worked

Reasoning

The model is now driven by business process and grain rather than by whatever relationships Power BI auto-detects. Professional upgrade: Professional modeling creates a reusable analytical contract. It anticipates report authors, refresh behavior, security, future measures and validation-not only today's visual.

10. Common beginner mistakes

1. Starting with charts. | 2. Creating relationships because matching column names exist. | 3. Mixing multiple grains in one fact without deciding what one row means. | 4. Treating a PBIX opening without errors as validation.

11. If you get stuck - recovery ladder

Recovery

Close Model view mentally. Write: process, one fact row means, questions, dimensions, controls. If any field is unclear, resolve it before modeling. Save your genuine model/design attempt before review.

12. Now you do a small version alone

Fresh independent task

Fresh case: parcel delivery. Define a model contract for shipment events, delivery charges, customers, hubs and dates.

13. Validate your result

Validation

Every requested business question maps to a fact at the correct grain and a filtering dimension; at least four independent controls are listed.

14. Teach it back in your own words

Explain without reading

Explain why a semantic model is not just a group of imported tables.

15. Protected review + changed fresh retry

Attempt protection

Save the model contract first. Review only after attempt; then redesign a changed support-operations scenario.

16. Evidence / mastery + exact next route

Mastery evidence

Model contract + sketch + validation plan + 60-90 second explain-back. VIDEO SUPPORT: 00:43-18:07
https://learn.microsoft.com/en-us/power-bi/guidance/star-schema After saving evidence, continue to PBM-A2.

PBM-A2 - Fact and dimension tables

1. Why it matters

A table is not a fact or dimension because of its filename. The analytical role depends on grain and relationships. If the 'one' side is not unique, a simple relationship can become ambiguous or invalid.

2. What you are learning

Classify facts versus dimensions by business role; identify fact grain and dimension key; test uniqueness/nonblank keys; distinguish measures/events from descriptive attributes.

3. Prerequisite / what you already know

PBM-A1 model contract.

4. Picture it in your head

Mental model

Imagine this as a visible path: Process event -> Fact -> Keys -> Dimension -> Filter/group -> Aggregate. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Fact table: A table representing measurable business events or states, usually on the many side of relationships. Dimension table: A table describing business entities used to filter/group facts, usually on the one side. Surrogate key: A technical key created to identify dimension rows when business keys are unsuitable or need historical versions.

6. Watch / visual source

Role/grain/key table + key-quality checks + explain-back. VIDEO: intentionally book-led for this lesson
https://learn.microsoft.com/en-us/power-bi/guidance/star-schema
After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. FactSales: one row per order line; columns include ProductID, CustomerID, OrderDate, Quantity and LineAmount.
2. DimProduct: one row per ProductID; descriptive attributes include Category, Brand and ProductName.
3. DimCustomer: one row per CustomerID. Before relationship creation, prove CustomerID is unique and nonblank.
4. MonthlyTargets is not a normal dimension just because it has Region. It is a higher-grain fact: one row per Region-Month target.
5. Do not mix Product descriptions into FactSales merely because it feels convenient; decide model role and storage trade-offs deliberately.

8. What you should see / be able to say

Expected result

A table-role register with grain, candidate key, one/many role and key-quality tests.

9. Why it worked

Reasoning

Each table has one analytical responsibility and relationship assumptions are testable. Professional upgrade: Dimension and fact roles are semantic. Size can vary. A small event fact is still a fact; a very large customer dimension is still a dimension if its relationship role fits.

10. Common beginner mistakes

1. Calling every lookup-like table a dimension without testing uniqueness. | 2. Putting subtotal rows inside a fact. | 3. Treating targets as a dimension. | 4. Creating a one-to-many relationship before checking the 'one' key.

11. If you get stuck - recovery ladder

Recovery

For each table say 'one row means...'. Then mark columns as keys/descriptions/measures. Test the supposed dimension key. Save your genuine model/design attempt before review.

12. Now you do a small version alone

Fresh independent task

Fresh case: classify Ticket, TicketEvent, Agent, TeamTarget and Customer tables and state each grain.

13. Validate your result

Validation

Dimension-side keys are unique/nonblank or the design records how they will be repaired.

14. Teach it back in your own words

Explain without reading

Explain fact versus dimension without saying 'big table versus small table'.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry classifies clinic encounters, billing events and provider dimensions.

16. Evidence / mastery + exact next route

Mastery evidence

Role/grain/key table + key-quality checks + explain-back. VIDEO: intentionally book-led for this lesson https://learn.microsoft.com/en-us/power-bi/guidance/star-schema After saving evidence, continue to PBM-A3.

PBM-A3 - Star schema in professional BI

1. Why it matters

Database source schemas are often normalized for transactions, not analytics. A clean star reduces ambiguous paths and makes filters/measures easier for report authors to understand.

2. What you are learning

Arrange conformed dimensions around facts, avoid fact-to-fact shortcuts, handle multiple facts deliberately, and recognize when a snowflake or higher-grain fact needs transformation.

3. Prerequisite / what you already know

PBM-A2 facts/dimensions.

4. Picture it in your head

Mental model

Imagine this as a visible path: Dimensions -> -> -> Fact -> <- -> Dimensions -> Measures. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Star schema: A model with fact tables surrounded by dimensions connected through clear one-to-many relationships. Conformed dimension: A dimension whose meaning/key can consistently filter multiple fact tables. Snowflake: A dimension split into related normalized tables rather than flattened into one analytical dimension.

6. Watch / visual source

Star diagram + fact-grain notes + filter-path review + explain-back. VIDEO SUPPORT: 02:18-15:35
https://www.youtube.com/watch?v=4ePNrdxWtY0
https://learn.microsoft.com/en-us/power-bi/guidance/star-schema
After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Put DimDate, DimProduct and DimCustomer around FactSales.
2. FactReturns stays a separate fact at return-event grain. Relate it through compatible dimensions or a deliberate business key strategy; do not create a casual fact-to-fact relationship.
3. MonthlyTargets stays a higher-grain fact connected only to dimensions whose grain it supports, such as Month/Region.
4. Use conformed dimensions so Sales and Returns interpret Product/Customer/Date consistently.
5. Sketch filter paths before building. Every important visual should have one understandable route from dimension to fact.

8. What you should see / be able to say

Expected result

A star-schema sketch showing facts at their own grains and shared dimensions without ambiguous fact-to-fact shortcuts.

9. Why it worked

Reasoning

Dimensions become the reusable filter surface; facts remain separate measurable processes. Professional upgrade: Good star schemas balance usability, performance and source constraints. Deviations are allowed when justified, but the default should be easy to reason about.

10. Common beginner mistakes

1. One giant flat table by default. | 2. Fact-to-fact many-to-many because both tables contain OrderID. | 3. Connecting targets to transaction rows without grain control. | 4. Snowflaking tiny descriptive hierarchies without benefit.

11. If you get stuck - recovery ladder

Recovery

Remove every relationship line. Reconnect only dimension-one to fact-many paths justified by grain. Recheck each business question. Save your genuine model/design attempt before review.

12. Now you do a small version alone

Fresh independent task

Fresh case: model Orders, Refunds and MonthlyMarketingTargets with Product, Customer, Channel and Date dimensions.

13. Validate your result

Validation

Trace at least five filter paths. No path should require guessing which relationship is intended.

14. Teach it back in your own words

Explain without reading

Explain why star schema is about predictable filtering, not drawing a star shape.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry uses support tickets + staffing targets.

16. Evidence / mastery + exact next route

Mastery evidence

Star diagram + fact-grain notes + filter-path review + explain-back. VIDEO SUPPORT: 02:18-15:35 https://learn.microsoft.com/en-us/power-bi/guidance/star-schema After saving evidence, continue to PBM-A4.

PBM-A4 - Relationships and filter propagation

1. Why it matters

A relationship is not merely a join line. It defines how filter context can propagate. The wrong active state or cross-filter direction can make visuals silently include/exclude the wrong rows.

2. What you are learning

Choose cardinality, active state and cross-filter direction deliberately; understand filter propagation from one-side dimensions to many-side facts; test behavior with a tiny matrix.

3. Prerequisite / what you already know

PBM-A3 star schema; basic Power BI Model view navigation.

4. Picture it in your head

Mental model

Imagine this as a visible path: Dimension filter -> Relationship -> Fact rows -> Measure -> Visual. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Filter propagation: How a filter on one model table affects rows in related tables. Cross-filter direction: The direction in which a relationship allows filters to travel. Active relationship: The default relationship path used for filter propagation.

6. Watch / visual source

Model screenshot + relationship contract + tiny filter test + total reconciliation + explain-back. VIDEO SUPPORT: 11:50-18:07
https://www.youtube.com/watch?v=4ePNrdxWtY0
https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-relationships-understand
After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Confirm DimProduct ProductID is unique and FactSales ProductID may repeat.
2. Create 1:* DimProduct -> FactSales on ProductID. Keep single-direction filtering from dimension to fact unless a specific design requires otherwise.
3. Build a tiny matrix: Product Category rows, Sales Amount measure. Select one category and predict which FactSales rows survive.
4. Temporarily inspect an unmatched ProductID case. Understand that relationship/data-integrity choices can change whether unknown facts appear.
5. Use Model view line/cardinality/filter indicators as evidence, then validate totals rather than trusting the icon alone.

8. What you should see / be able to say

Expected result

A relationship contract plus a small visual proving expected filter propagation.

9. Why it worked

Reasoning

The one-side dimension controls the filter and the fact supplies aggregation, matching star-schema reasoning. Professional upgrade: Prefer simple active dimension-to-fact paths. Bidirectional filtering is a design tool, not a default fix for a confusing model.

10. Common beginner mistakes

1. Using Both everywhere. | 2. Trusting auto-detected relationships. | 3. Creating active alternative paths that cause ambiguity. | 4. Testing only a grand total where filter problems are hidden.

11. If you get stuck - recovery ladder

Recovery

Use one dimension and one fact only. Check key uniqueness, relationship columns, cardinality and direction. Build a one-field matrix and predict before clicking. Save your genuine model/design attempt before review.

12. Now you do a small version alone

Fresh independent task

Fresh case: Team dimension filtering Ticket fact. Add a second table and prove whether the path remains unambiguous.

13. Validate your result

Validation

At least three slicer/matrix selections return expected filtered totals and grand total still reconciles.

14. Teach it back in your own words

Explain without reading

Explain filter propagation using 'which fact rows are allowed to remain'.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry uses Customer -> Orders with an alternative Region path.

16. Evidence / mastery + exact next route

Mastery evidence

Model screenshot + relationship contract + tiny filter test + total reconciliation + explain-back. VIDEO SUPPORT: 11:50-18:07 https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-relationships-understand After saving evidence, continue to PBM-A5.

PBM-A5 - Cardinality and many-to-many risks

1. Why it matters

Many-to-many can be legitimate, but it is also a common symptom of dirty keys or mixed grain. Changing cardinality to make an error disappear can create ambiguous or inflated analysis.

2. What you are learning

Diagnose duplicate one-side keys first; distinguish data-quality defect from genuine many-to-many business relationships; use a bridge or higher-grain fact design when justified.

3. Prerequisite / what you already know

PBM-A2 key quality and PBM-A4 relationships.

4. Picture it in your head

Mental model

Imagine this as a visible path: Duplicate key -> Why duplicate? -> True M:M? -> Bridge/higher gra... -> Validate. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Many-to-many relationship: A relationship where matching values can repeat on both sides. Bridge table: A table used to resolve a genuine many-to-many relationship by storing valid key combinations. Higher-grain fact: A fact table summarized at a broader level than another fact, such as Region-Month targets.

6. Watch / visual source

Diagnosis note + bridge/higher-grain sketch + key checks + filter test + explain-back. VIDEO: intentionally book-led for this lesson
https://learn.microsoft.com/en-us/power-bi/guidance/relationships-many-to-many
After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. DimSalesperson is one row per SalespersonID. DimRegion is one row per RegionID.
2. One salesperson can belong to several regions and one region can have several salespeople. This is a genuine many-to-many dimension relationship.
3. Create BridgeSalespersonRegion with one row per valid salesperson-region assignment. Connect each dimension 1:* to the bridge.
4. Do not use this pattern to excuse duplicate ProductID rows in DimProduct. Duplicate product keys are a different problem.
5. Validate bridge uniqueness at the pair grain and test a simple salesperson/region selection for expected membership.

8. What you should see / be able to say

Expected result

A model decision that explains whether the repeated key is a defect, higher-grain fact or genuine many-to-many relationship.

9. Why it worked

Reasoning

The bridge stores the relationship itself rather than forcing ambiguous duplication into a dimension. Professional upgrade: Modeling many-to-many requires explicit business semantics. Use it because the domain is many-to-many, not because Power BI offers the option.

10. Common beginner mistakes

1. Switching to many-to-many when the dimension key is dirty. | 2. Using bidirectional filters as the first repair. | 3. Joining monthly targets directly to every transaction row. | 4. Building a bridge with duplicate key pairs.

11. If you get stuck - recovery ladder

Recovery

Ask why each key repeats. If it should not repeat, repair the dimension. If repetition represents a real relationship, state bridge grain and test pair uniqueness. Save your genuine model/design attempt before review.

12. Now you do a small version alone

Fresh independent task

Fresh case: consultants can support several practices and practices have several consultants. Design the bridge and filter paths.

13. Validate your result

Validation

Bridge pair key is unique; dimensions remain one-side; selections produce expected membership without fact duplication.

14. Teach it back in your own words

Explain without reading

Explain the difference between a dirty duplicate key and a true many-to-many relationship.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry handles higher-grain region-month targets instead of a bridge.

16. Evidence / mastery + exact next route

Mastery evidence

Diagnosis note + bridge/higher-grain sketch + key checks + filter test + explain-back. VIDEO: intentionally book-led for this lesson https://learn.microsoft.com/en-us/power-bi/guidance/relationships-many-to-many After saving evidence, continue to PBM-A6.

PBM-A6 - Role-playing dimensions

1. Why it matters

The same entity can play different roles. A Sales row may have OrderDate and ShipDate; a Flight may have DepartureAirport and ArrivalAirport. One generic relationship cannot always express both reporting needs cleanly.

2. What you are learning

Recognize role-playing dimensions; choose duplicated role dimensions when simultaneous filtering is needed; understand active/inactive alternatives and USERELATIONSHIP conceptually.

3. Prerequisite / what you already know

PBM-A4 relationship behavior. DAX USERELATIONSHIP syntax will be taught more deeply in C2-04.

4. Picture it in your head

Mental model

Imagine this as a visible path: One entity -> Multiple roles -> Active default -> Alternative role -> Report need. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Role-playing dimension: One business entity used in multiple roles relative to a fact. Inactive relationship: A model relationship that does not propagate filters by default. USERELATIONSHIP: A DAX function that activates an existing relationship for a calculation.

6. Watch / visual source

Role requirement matrix + model sketch + filter tests + explain-back. VIDEO: intentionally book-led for this lesson
https://learn.microsoft.com/en-us/power-bi/guidance/relationships-active-inactive
After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. FactSales contains OrderDate and ShipDate. Ask whether reports must filter/group both date roles simultaneously.
2. If order-date analysis is dominant and ship-date is only needed in specific measures, one Date table with OrderDate active and ShipDate inactive can be acceptable.
3. If report authors must independently slice Order Date and Ship Date at the same time, duplicate role dimensions such as Order Date and Ship Date are clearer.
4. Name role tables/columns clearly so visuals do not show ambiguous generic 'Year' labels.
5. Validate both role paths with tiny visuals before building a dashboard.

8. What you should see / be able to say

Expected result

A documented role-playing decision driven by report requirements, not by convenience.

9. Why it worked

Reasoning

The design starts from how users need to filter, making relationship roles explicit. Professional upgrade: Microsoft guidance generally favors active relationships where possible; duplicated role dimensions often improve usability when independent filtering is required.

10. Common beginner mistakes

1. Assuming one Date table can be actively related to every date column at once. | 2. Duplicating dimensions without renaming role-specific fields. | 3. Using inactive relationships without documenting the measure behavior. | 4. Ignoring RLS implications of inactive paths.

11. If you get stuck - recovery ladder

Recovery

List each role and one report question using it. Decide whether simultaneous filtering is required. Then select active/inactive or duplicated-role design. Save your genuine model/design attempt before review.

12. Now you do a small version alone

Fresh independent task

Fresh case: Ticket has CreatedDate, ResolvedDate and SLADeadlineDate. Choose the role design and justify it.

13. Validate your result

Validation

Each required report question has an unambiguous filter path; field names communicate role.

14. Teach it back in your own words

Explain without reading

Explain a role-playing dimension using one person acting as customer and referrer in different contexts.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry uses Departure and Arrival airport roles.

16. Evidence / mastery + exact next route

Mastery evidence

Role requirement matrix + model sketch + filter tests + explain-back. VIDEO: intentionally book-led for this lesson https://learn.microsoft.com/en-us/power-bi/guidance/relationships-active-inactive After saving evidence, continue to PBM-A7.

PBM-A7 - Date dimensions

1. Why it matters

Hidden auto-date behavior can be convenient for quick work but becomes harder to govern in reusable BI models. A deliberate calendar gives consistent fiscal/month/week attributes and controlled relationships.

2. What you are learning

Design a continuous date dimension, choose start/end range, add reusable attributes, keep one row per date, mark/use the date table when required, and connect date roles intentionally.

3. Prerequisite / what you already know

PBM-A6 date roles and basic date fields.

4. Picture it in your head

Mental model

Imagine this as a visible path: Continuous dates -> Attributes -> One-side key -> Fact date roles -> Time measures. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Date dimension: A table containing one row per calendar date plus attributes used for filtering/grouping. Continuous date range: Every date between model start and end exists with no gaps. Auto date/time: Power BI behavior that can create hidden date tables/hierarchies for date columns.

6. Watch / visual source

Date-table specification + continuity tests + role relationships + explain-back. VIDEO: intentionally book-led for this lesson
https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-date-tables
After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Set range from the minimum required business date to maximum reporting horizon; include every day, not only days with sales.
2. Columns may include Date, Year, MonthNumber, MonthName, YearMonth, Quarter, Week and fiscal attributes only when the business uses them.
3. Prove Date is unique, nonblank and continuous. Sort MonthName by MonthNumber where appropriate.
4. Relate DimDate to the primary fact date. Document any role-playing alternatives from PBM-A6.
5. Validate that a month with zero events can still appear when the report needs calendar continuity.

8. What you should see / be able to say

Expected result

One governed calendar table with continuous unique Date key and reusable attributes.

9. Why it worked

Reasoning

Time grouping comes from one consistent dimension rather than independent hidden date hierarchies for each column. Professional upgrade: A date dimension is shared analytical infrastructure. Its fiscal and period definitions should be governed like KPI definitions.

10. Common beginner mistakes

1. Creating the calendar only from dates that exist in the fact. | 2. Using MonthName without sort logic. | 3. Mixing fiscal and calendar definitions silently. | 4. Marking/using a date table without checking uniqueness/continuity.

11. If you get stuck - recovery ladder

Recovery

Test MIN/MAX, COUNTROWS versus date difference + 1, duplicate Date count, blank Date count. Repair the calendar before relationships. Save your genuine model/design attempt before review.

12. Now you do a small version alone

Fresh independent task

Fresh case: build a fiscal calendar contract for Apr-Mar reporting with one row per date.

13. Validate your result

Validation

Unique/nonblank/continuous Date key; attributes reproduce expected fiscal boundaries; zero-event dates can exist.

14. Teach it back in your own words

Explain without reading

Explain why a date table contains dates where nothing happened.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry creates separate role labels for Created Date and Closed Date.

16. Evidence / mastery + exact next route

Mastery evidence

Date-table specification + continuity tests + role relationships + explain-back. VIDEO: intentionally book-led for this lesson https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-date-tables After saving evidence, continue to PBM-A8.

PBM-A8 - Power Query architecture for BI models

1. Why it matters

A semantic model can be clean while its Power Query layer is fragile. Mixing source connection, cleaning, business logic and final load into one opaque query makes refresh/debugging harder.

2. What you are learning

Separate source/staging/model-ready queries; set types deliberately; reuse references; document transformations; understand when query folding can push work to a capable source.

3. Prerequisite / what you already know

C2-01 Power Query concepts if you took C2A sequentially. This lesson restates the BI-model-specific architecture from first principles.

4. Picture it in your head

Mental model

Imagine this as a visible path: Source -> Staging -> Reusable transform -> Model-ready -> Load -> Refresh. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Staging query: A query used to establish source scope/types/basic cleaning before downstream model-ready queries. Query folding: Power Query translating supported transformations into operations executed by the source system. Reference query: A new query based on another query's result/steps, useful for modular design.

6. Watch / visual source

Query dependency sketch + Applied Steps evidence + folding note + row/key controls + explain-back. VIDEO SUPPORT: 04:54-13:36; 17:50-28:25; 33:35-35:17
https://www.youtube.com/watch?v=MMdcczmULrU
https://learn.microsoft.com/en-us/power-query/best-practices
After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. qSales_Source connects and establishes source scope. Keep source-specific details here.
2. qSales_Staging sets types, standardizes fields and performs low-risk cleaning. Record row-count changes.
3. Reference staging to create FactSales_Model and any justified dimensions rather than duplicating source logic.
4. For relational sources, inspect folding availability where supported; do not assume CSV/Excel file sources fold.
5. Disable load for intermediate queries when appropriate and validate final model-ready row/key controls before loading.

8. What you should see / be able to say

Expected result

A query dependency design where source, staging and model-ready responsibilities are obvious and refresh behavior can be diagnosed.

9. Why it worked

Reasoning

Modular queries separate concerns and reduce duplicated transformation logic. Professional upgrade: Current Microsoft guidance emphasizes types, early filters, modular queries, documentation, future-proofing and connector-aware folding.

10. Common beginner mistakes

1. One giant query for everything. | 2. Copying queries instead of referencing shared logic without reason. | 3. Claiming query folding on CSV/Excel files. | 4. Optimizing folding before checking correctness.

11. If you get stuck - recovery ladder

Recovery

Map query dependencies. Identify source, reusable cleaning and final model outputs. Split only where responsibilities become clearer. Save your genuine model/design attempt before review.

12. Now you do a small version alone

Fresh independent task

Fresh case: design staging/model-ready queries for shipment CSV + SQL customer master; state which source might support folding and why.

13. Validate your result

Validation

Every loaded table has an upstream path, row/key controls and an intentional load setting.

14. Teach it back in your own words

Explain without reading

Explain query folding as 'send supported work back to the data source' rather than 'Power Query is faster'.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry reorganizes an opaque five-step query family into source/staging/model-ready groups.

16. Evidence / mastery + exact next route

Mastery evidence

Query dependency sketch + Applied Steps evidence + folding note + row/key controls + explain-back. VIDEO SUPPORT: 04:54-13:36; 17:50-28:25; 33:35-35:17 https://learn.microsoft.com/en-us/power-query/best-practices After saving evidence, continue to PBM-A9.

PBM-A9 - Model validation and reconciliation

1. Why it matters

A model can pass relationship creation and still understate unknown products, duplicate values through a bridge, or filter through an unintended path. Validation must prove model behavior, not only table totals.

2. What you are learning

Create a model QA pack covering source row/totals, dimension key quality, unmatched keys, relationship cardinality/active/direction, filter-propagation tests, measure reconciliation and role/date behavior.

3. Prerequisite / what you already know

PBM-A1-A8 and ability to create basic Power BI visuals.

4. Picture it in your head

Mental model

Imagine this as a visible path: Source controls -> Keys -> Relationships -> Filter tests -> Measures -> Sign-off. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Model reconciliation: Comparing model outputs to independent source/control totals. Unknown member: A deliberate dimension row used in some designs to retain facts whose dimension key is missing/unmatched. Propagation test: A small visual/slicer test proving filters travel through intended relationships.

6. Watch / visual source

Model QA pack + exception list + propagation screenshots + reconciliation + explain-back. VIDEO: intentionally book-led for this lesson
https://learn.microsoft.com/en-us/power-bi/guidance/relationships-troubleshoot
After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Record source FactSales rows and NetSales control before Power BI transformations.
2. Prove dimension keys are unique/nonblank and list unmatched ProductID/CustomerID facts.
3. Record each relationship: columns, cardinality, active state and cross-filter direction.
4. Build a tiny matrix per dimension and compare selected/grand totals to an independent control.
5. Test Order Date versus Ship Date role behavior. Record any expected blank/unknown handling and sign off only after discrepancies are explained.

8. What you should see / be able to say

Expected result

A QA workbook/log that proves both data and relationship behavior before report sign-off.

9. Why it worked

Reasoning

Controls cover distinct failure classes: source loss, key defect, relationship path and calculation output. Professional upgrade: Save model QA evidence with releases. When a source/model change occurs, rerun the affected control set rather than relying on visual familiarity.

10. Common beginner mistakes

1. Only comparing one grand total. | 2. Fixing unmatched keys by filtering them out silently. | 3. Testing relationships with a complex dashboard instead of a tiny visual. | 4. Treating a green refresh as validation.

11. If you get stuck - recovery ladder

Recovery

Reduce the model to one dimension + one fact. Reconcile source rows/total, key uniqueness, unmatched list, then add one relationship/filter test at a time. Save your genuine model/design attempt before review.

12. Now you do a small version alone

Fresh independent task

Fresh case: validate a shipment model with one unknown hub and two date roles.

13. Validate your result

Validation

Every material mismatch has cause, owner and disposition; no critical issue is hidden to make totals match.

14. Teach it back in your own words

Explain without reading

Explain why model validation needs both data controls and filter-behavior controls.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry validates a clinic model with a many-to-many bridge.

16. Evidence / mastery + exact next route

Mastery evidence

Model QA pack + exception list + propagation screenshots + reconciliation + explain-back. VIDEO: intentionally book-led for this lesson https://learn.microsoft.com/en-us/power-bi/guidance/relationships-troubleshoot After saving evidence, continue to PBM-A10.

PBM-A10 - Model performance fundamentals

1. Why it matters

Slow reports tempt learners to rewrite DAX randomly. Performance often begins with model shape: unnecessary high-cardinality columns, poor relationship design, over-wide imports or non-folding transformations.

2. What you are learning

Diagnose before optimizing; remove unused columns, reduce avoidable cardinality, preserve star design, use Performance Analyzer for observed bottlenecks, and measure one change at a time.

3. Prerequisite / what you already know

PBM-A1-A9. Advanced DAX optimization comes later in C2-04.

4. Picture it in your head

Mental model

Imagine this as a visible path: Measure first -> Model shape -> Cardinality -> Relationships -> Analyzer -> Change one thing. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Cardinality level: How many distinct values a column contains; high cardinality can increase storage and processing cost. Performance Analyzer: Power BI tool for capturing timing information for visual interactions. Aggregation: A summarized table/model strategy that can accelerate queries on large data sets in appropriate designs.

6. Watch / visual source

Before/after model inventory + performance evidence where available + result parity + explain-back. VIDEO: intentionally book-led for this lesson
https://learn.microsoft.com/en-us/training/modules/optimize-semantic-model-performance/
After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. First confirm the report result is correct. Performance work cannot compensate for wrong semantics.
2. Inspect model: remove unused verbose text columns from FactSales; keep descriptive text in dimensions where appropriate.
3. Check relationship complexity and avoid unnecessary bidirectional/ambiguous paths.
4. Use Performance Analyzer to capture a slow interaction and identify where time is spent rather than guessing.
5. Change one model/query/report factor, rerun the same interaction and record before/after evidence.

8. What you should see / be able to say

Expected result

A performance note linking one measured symptom to one justified change while preserving result parity.

9. Why it worked

Reasoning

The diagnosis starts from observable evidence and model design before advanced formula tricks. Professional upgrade: Current Microsoft training emphasizes Performance Analyzer, DAX optimization, cardinality reduction and aggregations. C2-03 focuses on the model side first; C2-04 deepens DAX.

10. Common beginner mistakes

1. Deleting columns without checking report dependencies. | 2. Assuming high row count alone is the problem. | 3. Using Both relationships to 'fix' slow/incorrect visuals. | 4. Making five changes before measuring again.

11. If you get stuck - recovery ladder

Recovery

Restore known-good model, capture baseline, inspect model width/cardinality/relationships, change one thing and retest. Save your genuine model/design attempt before review.

12. Now you do a small version alone

Fresh independent task

Fresh case: a report includes transaction GUID, long comments and duplicated descriptive columns in a large fact. Propose and test a cleanup plan.

13. Validate your result

Validation

Key report totals remain identical; removed columns are unused; analyzer/baseline evidence records the measured change where runtime is available.

14. Teach it back in your own words

Explain without reading

Explain why 'fewer columns' can matter more than a clever DAX rewrite.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry diagnoses a different model with a relationship problem rather than wide text columns.

16. Evidence / mastery + exact next route

Mastery evidence

Before/after model inventory + performance evidence where available + result parity + explain-back. VIDEO: intentionally book-led for this lesson https://learn.microsoft.com/en-us/training/modules/optimize-semantic-model-performance/ After saving evidence, complete the module mastery checklist and follow the Gate A / completion route.

C2-03 mastery checklist + quick reference

Use after the first full pass; do not replace Desktop practice with reading

TEN MODEL QUESTIONS

A1 What process/questions/grain does the model contract support? | A2 Which tables are facts/dimensions and are one-side keys clean? | A3 Are facts separated and dimensions conformed? | A4 Which direction should filters propagate? | A5 Is repetition a defect, higher-grain fact or true M:M? | A6 Which entity roles need separate active dimensions? | A7 Is Date unique/continuous/governed? | A8 Are source/staging/model-ready query responsibilities clear? | A9 Do model/filter results reconcile? | A10 What measured evidence justifies optimization?

BLOCK MASTERY IF

Fact grain mixed or unstated; one-side key duplicated; M:M used as a shortcut; ambiguous/bidirectional path unexplained; role/date filter wrong; unknown facts silently dropped; source/model totals do not reconcile; performance claims made without observed evidence.

FAST MODEL DEBUG

Source controls -> one-side key quality -> unmatched keys -> relationship columns/cardinality/active/direction -> tiny slicer/matrix propagation test -> measure total -> role/date tests -> only then dashboard complexity.

COURSE POSITIONING

C2-03 is optional BI depth. Its modeling skill is valuable for Senior/BI roles; direct DA-to-DE learners may take C2B instead, where required modeling competencies are taught for the engineering handoff.

C2-03 source / video registry

Current Microsoft modeling guidance + two preserved visual companions

PBM-A1/A2/A3 - Star schema guidance

https://learn.microsoft.com/en-us/power-bi/guidance/star-schema

Chandoo modeling video: 00:43-18:07

PBM-A4 - Model relationships in Power BI Desktop

https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-relationships-understand

Chandoo 11:50-18:07

PBM-A5 - Many-to-many relationship guidance

https://learn.microsoft.com/en-us/power-bi/guidance/relationships-many-to-many

Book-led

PBM-A6 - Active vs inactive relationship guidance

https://learn.microsoft.com/en-us/power-bi/guidance/relationships-active-inactive

Book-led

PBM-A7 - Set and use date tables

https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-date-tables

Book-led

PBM-A8 - Power Query best practices

https://learn.microsoft.com/en-us/power-query/best-practices

Chandoo PQ 04:54-13:36; 17:50-28:25; 33:35-35:17

PBM-A8 - Query folding basics

https://learn.microsoft.com/en-us/power-query/query-folding-basics

Current connector-aware folding reference

PBM-A9 - Relationship troubleshooting

https://learn.microsoft.com/en-us/power-bi/guidance/relationships-troubleshoot

Book-led

PBM-A10 - Optimize semantic model performance

https://learn.microsoft.com/en-us/training/modules/optimize-semantic-model-performance/

Performance Analyzer/cardinality/aggregations

VERIFICATION RULE

Official/current references and preserved visual companions were rechecked on the web 2026-09-10. External media is support only. Power BI Desktop execution and the internal model QA pack remain the mastery evidence.